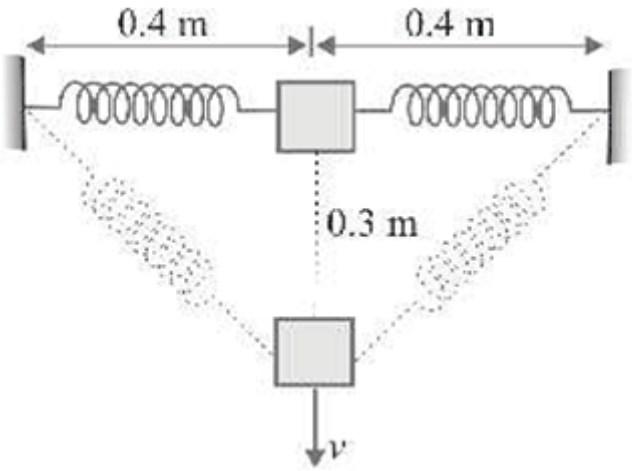
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In figure, a block having a mass of $5\ \text{kg}$ is released from rest, where the spring acting on the body are horizontal and have a tension of $50\ \text{N}$. The force constant of each spring is, $k = 500\ \text{N/m}$. The velocity of the block after it has descended
by $0.3\ \text{m}$.



(A) 0

(B) $1\ \text{m/s}$

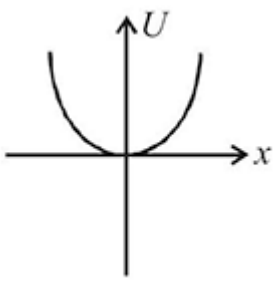
(C) $2\ \text{m/s}$

(D) none

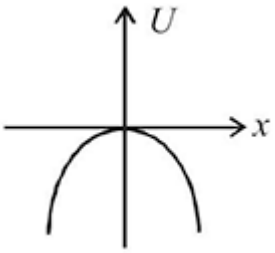
Q.2

A particle is acted by a force $F = kx$, where k is a +ve constant. Its potential energy at $x = 0$ is zero. Which curve correctly represents the variation of potential energy of the block with respect to x

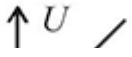
(A)



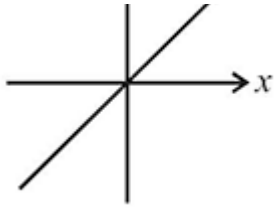
(B)



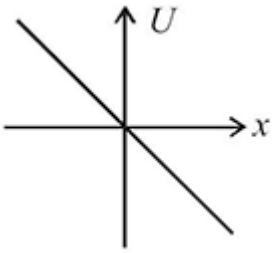
(C)



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(D)



Q.3

A moving body with a mass m_1 and velocity u strikes a stationary body of mass m_2. The masses m_1 and m_2 should be in the ratio m_1/m_2 so as to decrease the velocity of the first body to $2u/3$ and giving a velocity of v to m_2 assuming a perfectly elastic impact. Then the ratio m_1/m_2 is

(A) 5

(B) $1/5$

(C) $1/25$

(D) 25

Q.4

A body is allowed to fall freely under gravity from a height of $10\ m$. If it looses 25% of its energy due to impact with the ground, then the maximum height it rises after one impact is

(A) $2.5\ m$

(B) $5.0\ m$

(C) $7.5\ m$

(D) $8.2\ m$

Q.5

A glass marble dropped from a certain height above the horizontal surface reaches the surface in time t and then continues to bounce up and down. The time in which the marble finally comes to rest is

(A) $e^n t$

(B) $e^2 t$

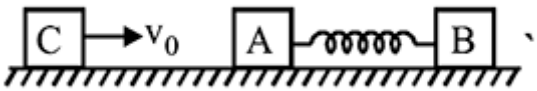
(C) $t\left[\frac{1+e}{1-e}\right]$

(D) $t\left[\frac{1-e}{1+e}\right]$

Q.6

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A block C of mass m is moving with velocity v_0 and collides elastically with block A of mass m and connected to another block B of mass $2\ m$ through spring constant k. What is k if x_0 is compression of spring when velocity of A and B is same?



(A) $\frac{mv_0^2}{x_0^2}$

(B) $\frac{mv_0^2}{2x_0^2}$

(C) $\frac{3}{2}\frac{mv_0^2}{x_0^2}$

(D) $\frac{2}{3}\frac{mv_0^2}{x_0^2}$

Q.7

In the figure shown, a particle of mass m is released from the position A on a smooth track. When the particle reaches at B, then normal reaction on it by the track is



(A) mg

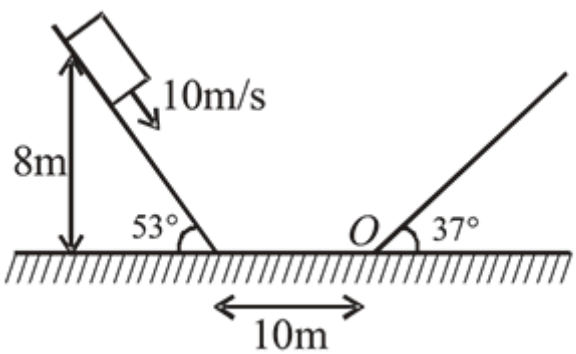
(B) $2mg$

(C) $\frac{2}{3}mg$

(D) $\frac{m^2\ g}{h}$

Q.8

A $1\ kg$ particle at a height of $8\ m$ has a speed of $10\ m/s$ down a fixed incline making an angle 53° with horizontal as shown in figure. It slides on a horizontal section of length $10\ m$ at ground level and then up a fixed incline making an angle 37° with horizontal. All surfaces have $\mu_k = 0.5$. How far (in meters) from point O (bottom of right inclined plane), along the incline making an angle 37° with horizontal, does the particle first come to rest?



(A) $1\ m$

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(B) $0.5\ \mathrm{m}$

(C) $2\ \mathrm{m}$

(D) $5\ \mathrm{m}$

Q.9

A vertical spring with force constant k is fixed on a table. A ball of mass m at a height h above the free upper end of the spring falls vertically on the spring so that the spring is compressed by a distance d. The net work done in the process is

(A) $\mathrm{mg(h+d)-\frac{1}{2}kd^2}$

(B) $mg(h-d)-\frac{1}{2}kd^2$

(C) $mg(h-d)+\frac{1}{2}kd^2$

(D) $\mathrm{mg(h+d)+\frac{1}{2}kd^2}$

Q.10

A $2\ \mathrm{kg}$ block slides on a horizontal floor with a speed of $4\ \mathrm{m/s}$. It strikes a uncompressed spring, and compresses it till the block is motionless. The kinetic friction force is $15\ \mathrm{N}$ and spring constant is $10,000\ \mathrm{N/m}$. The spring compresses (in cm) by

(A) 5cm

(B) 4.5cm

(C) 5.5cm

(D) 3cm

Q.11

A steel ball of mass $5\ \mathrm{g}$ is thrown downward with velocity $10\ \mathrm{m/s}$ from height $19.5\ \mathrm{m}$. It penetrates sand by $50\ \mathrm{cm}$. The change in mechanical energy (in joule) will be $\left(\mathrm{g}=10\ \mathrm{m/s^2}\right)$

(A) 1.25J

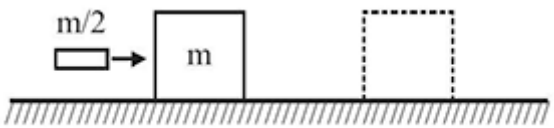
(B) 2J

(C) 1.75J

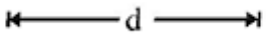
(D) 12.5J

Q.12

A block of mass m rests on a rough horizontal surface (Coefficient of friction is μ). When a bullet of mass $\mathrm{m/2}$ strikes horizontally, and get embedded in it, the block moves a distance d before coming to rest. The initial velocity of the bullet is



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(A) $3\sqrt{\mu g d}$

(B) $\sqrt{2\mu g d}$

(C) $2\sqrt{\mu g d}$

(D) $3\sqrt{2\mu g d}$

Q.13

Two blocks of masses $m_1 = 10\ \text{kg}$ and $m_2 = 20\ \text{kg}$ are connected by a spring of stiffness $k = 200\ \text{N/m}$. The coefficient of friction between the blocks and the fixed horizontal surface is $\mu = 0.1$. Find the minimum constant horizontal force F (in newtons) to be applied to m_1 in order to slide the mass m_2 [Take $g = 10\ \text{m/s}^2$]

(A) $\mu m_1 g + \frac{\mu m_2 g}{2}$

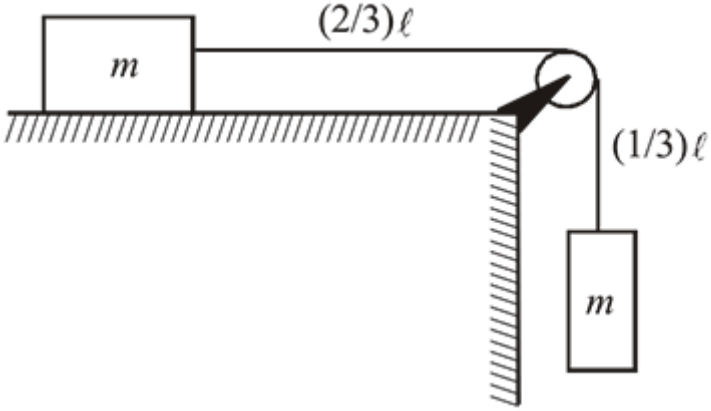
(B) $\mu m_1 g + \mu m_2 g$

(C) $\mu m_1 g - \frac{\mu m_2 g}{2}$

(D) $\frac{\mu m_1 g + \mu m_2 g}{2}$

Q.14

Two objects each of mass m are connected with a uniform rope having same mass m and length $= 10/3m$ are shown in the figure. The rope can slide on a frictionless pulley which is fixed to the edge of the table. Initially $1/3$ of the rope hangs vertically. Friction is negligible everywhere.
If the system is released from rest and object on the right hand side descend further by a distance of $\ell/3$, what would be the speed of the object (in m/s) ?



(A) $2.12\ \text{m/s}$

(B) $3.33\ \text{m/s}$

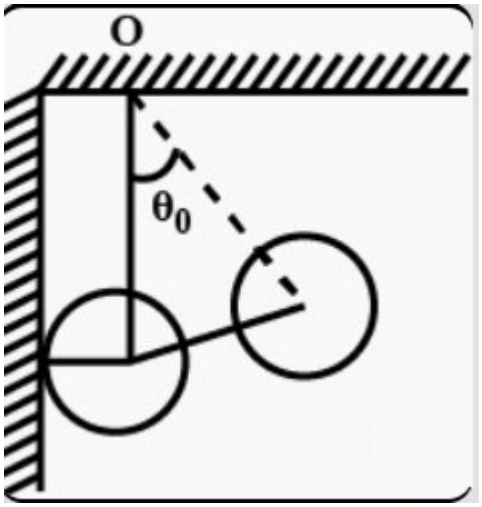
(C) $0.33\ \text{m/s}$

(D) $4.12\ \text{m/s}$

Q.15

An iron ball of mass m, suspended by a light inextensible string of length ℓ from a fixed point O, is shifted by an angle θ_0 as shown so as to strike the vertical wall perpendicularly. The maximum angle made by the string with vertical after the first collision, if e is the coefficient of restitution, is

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(A) $\sin^{-1}\left\{1-e^2\left(1-\cos\theta_0\right)\right\}$

(B) $\cos^{-1}\left\{1-e^2\left(1-\cos\theta_0\right)\right\}$

(C) $\tan^{-1}\left\{1-e^2\left(1-\cos\theta_0\right)\right\}$

(D) zero

Q.16

A particle of mass $10\ g$ moves along a circle of radius $6.4\ cm$ with a constant tangential acceleration. What is the magnitude of this acceleration if the kinetic energy of the particle becomes equal to $8\times 10^{-4}\ J$ by the end of the second revolution after the beginning of the motion?

(A) $0.1\ m/s^2$

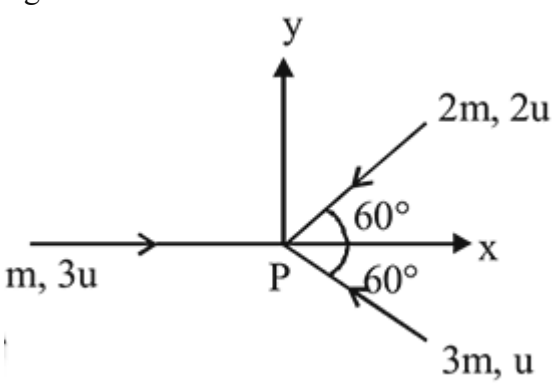
(B) $0.15\ m/s^2$

(C) $0.18\ m/s^2$

(D) $0.2\ m/s^2$

Q.17

Three masses $m, 2\ m$ and $3\ m$ are moving in $x-y$ plane with speed $3u, 2u$ and u respectively as shown in figure. The three masses collide at the same point at P and stick together. The velocity of resulting mass will be



(A) $\frac{u}{12}(\hat{i}+\sqrt{3}\hat{j})$

(B) $\frac{u}{12}(\hat{i}-\sqrt{3}\hat{j})$

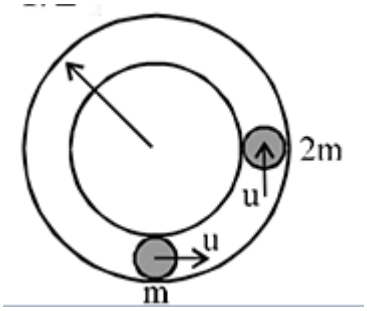
(C) $\frac{u}{12}(-\hat{i}+\sqrt{3}\hat{j})$

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(D) $\frac{u}{12}(-\hat{i}-\sqrt{3}\hat{j})$

Q.18

Two small bodies of masses 'm' and ' $2\ m$ ' are placed in a fixed smooth horizontal circular hollow tube of mean radius 'r' as shown. The mass ' m ' is moving with speed 'u' and the mass ' $2\ m$ ' is stationary. After their first collision, the time elapsed for next collision is $[$ coefficient of restitution $e=1/2]$



(A) $\frac{2\pi r}{u}$

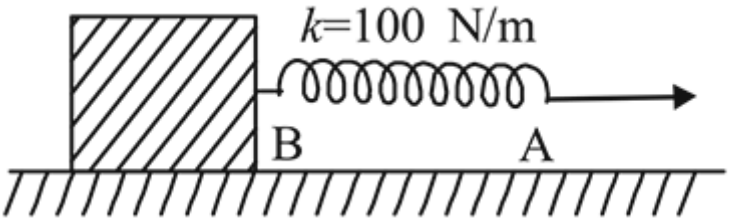
(B) $\frac{4\pi r}{u}$

(C) $\frac{3\pi r}{u}$

(D) $\frac{12\pi r}{u}$

Q.19

A block lying on a smooth surface with spring connected to it is pulled by an external force as shown. Initially the velocity of ends A and B of the spring are $4\ m/s$ and $2\ m/s$ respectively. If the energy of the spring is increasing at the rate of $20\ J/sec$, then the stretch in the spring is



(A) $1.0\ cm$

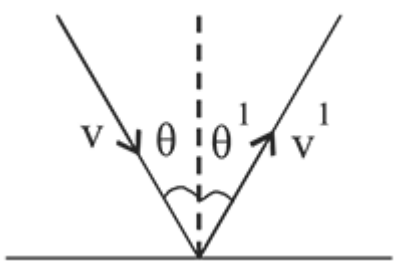
(B) $2.0\ cm$

(C) $10\ cm$

(D) $2.0\ cm$

Q.20

A ball of mass m hits the floor making an angle θ as shown in the figure. If e is the coefficient of restitution, then which relation is true, for the velocity component before and after collision?



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(A) $\mathrm{V}^1 \sin\theta' = \mathrm{V}\sin\theta$

(B) $\mathrm{V}^1 \sin\theta' = -\sin\theta$

(C) $\mathrm{V}^1 \cos\theta' = \mathrm{V}\cos\theta$

(D) . $\mathrm{V}^1 \cos\theta' = -\mathrm{V}\cos\theta$

Answers & Solutions

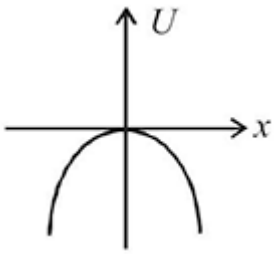
Q.1 Answer:

0

Solution:

$x_i = \frac{50}{500} = 0.1\ \mathrm{m}, \ell_i = 0.4 - 0.1 = 0.3\ \mathrm{m}$
$x_f = \sqrt{0.4^2 + 0.3^2} - 0.3 = 0.2\ \mathrm{m}$
$2 \times \frac{1}{2}kx_i^2 + mg \times 0.3 = 2 \times \frac{1}{2}kx_f^2 + \frac{1}{2}mv^2 \Rightarrow v = 0$

Q.2 Answer:



Solution:

We know that $\Delta\mathrm{U} = -\mathrm{W}$ **for conservative forces** $\Delta\mathrm{U} = -\int_0^x F dx$ **or** $\Delta\mathrm{U} = -\int_0^x kx dx$
$\Rightarrow \quad U_{(x)} - U_{(0)} = -\frac{kx^2}{2}$
Given $U_{(0)} = 0$
$U_{(x)} = -\frac{kx^2}{2}$
This is the equation of a parabola, which is symmetric to U-axis in negative direction.

Q.3 Answer:

5

Solution:

$\mathrm{m_1 u} = \mathrm{m_1}\frac{2\mathrm{u}}{3} + \mathrm{m_2 v}$ **(By condition of linear momentum)** $\Rightarrow \frac{1}{3}\ \mathrm{m_1 u} = \mathrm{m_2 v}$
. . .. . .
Also $\mathrm{e} = \frac{|\mathrm{v_1} - \mathrm{v_2}|}{|\mathrm{u_2} - \mathrm{u_1}|}$
$\Rightarrow \mathrm{v} - \frac{2\mathrm{u}}{3} = \mathrm{u} \Rightarrow \mathrm{v} = \frac{5}{3}\mathrm{u}$
From (i) and (ii), $\frac{1}{3}\ \mathrm{m_1 u} = \frac{5}{3}\ \mathrm{m_2 u} \Rightarrow \frac{\mathrm{m_1}}{\mathrm{m_2}} = 5$

Q.4 Answer:

7.5 m

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Solution:

Just before impact, energy $E = mgh = 10mg \quad \ldots(1)$

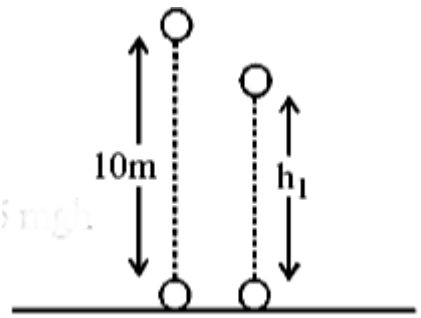
Just after impact

$E_1 = mgh - \frac{25}{100}mgh = 0.75mgh$

Hence, $mgh_1 = E_1$

(from given figure) $mgh_1 = 0.75mg(10)$

$h_1 = 7.5\ m$



Q.5 Answer:

$t\left[\frac{1+e}{1-e}\right]$

Solution:

$$t_{AB} = \sqrt{\frac{2h}{g}}$$

$$t_{BC} + t_{CB} = 2\sqrt{\frac{2h_1}{g}}$$

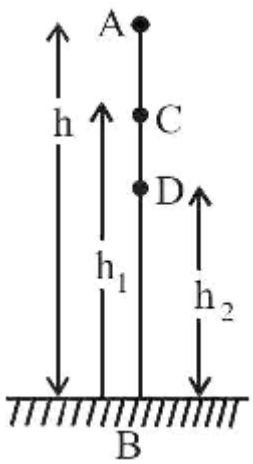
$$= 2\sqrt{\frac{2e^2 h}{g}} = 2e\sqrt{\frac{2h}{g}}$$

$$t_{BD} + t_{DB} = 2e^2\sqrt{\frac{2h}{g}}$$

- Total time taken by the body in coming to rest $= \sqrt{\frac{2h}{g}} + 2e\sqrt{\frac{2h}{g}} + 2e^2\sqrt{\frac{2h}{g}} + \ldots\ldots\ldots$

$= \sqrt{\frac{2h}{g}} + 2e\sqrt{\frac{2h}{g}}\left[1 + e + e^2 + \ldots\ldots\ldots\right]$

$= \sqrt{\frac{2h}{g}} + 2e\sqrt{\frac{2h}{g}} \times \frac{1}{1-e} = \sqrt{\frac{2h}{g}}\left[\frac{1+e}{1-e}\right] = t\left(\frac{1+e}{1-e}\right)$



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Q.6 Answer:

$\frac{2}{3}\frac{mv_0^2}{x_0^2}$

Solution:

When C strikes A $\frac{1}{2}mv_0{}^2 = \frac{1}{2}mv'^2 + \frac{1}{2}kx_0{}^2$ **(**$v' =$ **velocity of** A**)**

$kx_0^2 = m\left(v_0{}^2 - v'^2\right)$

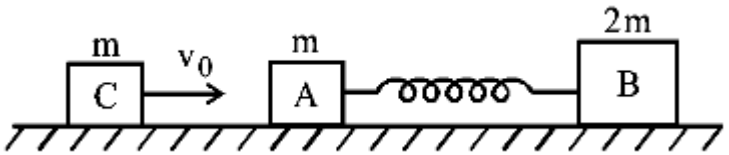
$\frac{1}{2}2mv'^2 = \frac{1}{2}kx_0{}^2$

(When A and B Block attains K.E.) $\therefore \quad \frac{1}{2}kx_0^2 = mv'^2$ **.....**

From (i) and (ii), $kx_0^2 = mv_0^2 - mv'^2 = mv_0^2 - \frac{k}{2}x_0{}^2$

$\Rightarrow kx_0^2 + \frac{k}{2}x_0^2 = mv_0^2$

$\frac{3}{2}kx_0^2 = mv_0^2 \therefore k = \frac{2}{3}\, m\frac{v_0^2}{x_0^2}$



Q.7 Answer:

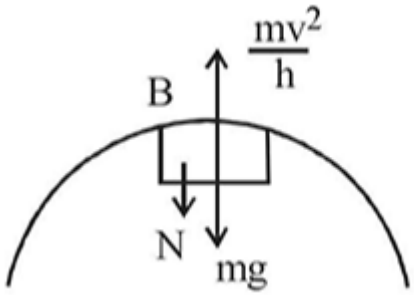
mg

Solution:

By conservation of energy $mg(3\ h) = mg(2\ h) + \frac{1}{2}mv^2 \quad (v =$ **velocity at** $B)$

$mgh = \frac{1}{2}mv^2; v = \sqrt{2gh}$

From free body diagram of block at B



$N + mg = \frac{mv^2}{h} = 2mg; N = mg$

Q.8 Answer:

$1\ m$

Solution:

Let x **be the distance travelled by the particle along the** 37° **incline, after which it comes to rest. From work-energy theorem, Work done by gravity on block on incline** $53^\circ +$ **work done by friction on block on incline** $53^\circ +$ **work done by friction on ground level + work done by gravity on block on incline** $37^\circ +$ **work done by friction on block on incline** $37^\circ =$ **change in** $K.E.\ mg(8) - \mu mg\cos 53^\circ(10) - \mu mg(10) - \mu mg\cos 37^\circ(x)$

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$-mg\sin 37^\circ(x) = 0 - \frac{1}{2}(1)(10)^2$

On solving we get $x = 1$ m

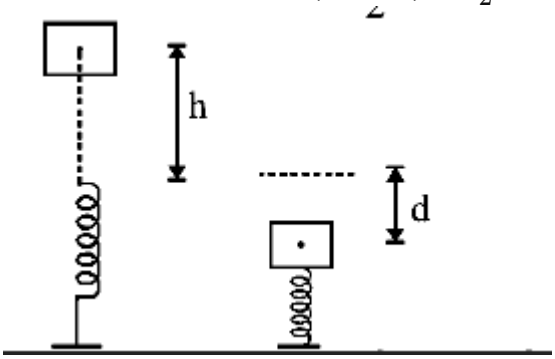
Q.9 Answer:

$\mathrm{mg(h+d)} - \frac{1}{2}\mathrm{kd}^2$

Solution:

Gravitational potential energy of ball gets converted into elastic potential energy of the spring. $\mathrm{mg(h+d)} = \frac{1}{2}\mathrm{kd}^2$

Net work done $= mg(h+d) - \frac{1}{2}kd^2 = 0$



Q.10 Answer:

5.5cm

Solution:

Let the block compress the spring by x **before stopping. Kinetic energy of the block** $=$ **(P.E of compressed spring)** $+$ **work done against friction.**

$\frac{1}{2} \times 2 \times (4)^2 = \frac{1}{2} \times 10,000 \times x^2 + 15 \times x$

$10,000x^2 + 30x - 32 = 0$

$\Rightarrow 5000x^2 + 15x - 16 = 0$

$\therefore x = \frac{-15 \pm \sqrt{(15)^2 - 4 \times (5000)(-16)}}{2 \times 5000}$

$= 0.055 \text{ m} = 5.5 \text{ cm}.$

Q.11 Answer:

1.25J

Solution:

$\mathrm{v}^2 = \mathrm{u}^2 + 2\mathrm{gh} = (10)^2 + 2 \times 10 \times 19.5 = 490$

K.E. at the ground

$= \frac{1}{2}\mathrm{mv}^2 = \frac{1}{2} \times \frac{5}{1000} \times 490 = \frac{49}{40}$ J

P.E. $= \mathrm{mgh} = \frac{5}{1000} \times 10 \times \left(\frac{-50}{100}\right) = -\frac{1}{40}$ J

$\therefore$ **Change in energy** $= \frac{49}{40} - \left(-\frac{1}{40}\right) = \frac{50}{40} = 1.25$ J

Q.12 Answer:

$3\sqrt{2\mu gd}$

Solution:

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Let initial velocity of the bullet be v.

By linear momentum conservation

$\frac{\mathrm{m}}{2}\mathrm{v} = \left(\frac{\mathrm{m}}{2} + \mathrm{m}\right)\mathrm{v}_1$

$(\mathrm{v}_1 =$ **combined velocity** $)$

$\mathrm{v}_1 = \frac{\mathrm{v}}{3}$

. . .. . .

retardation $= \mu g$

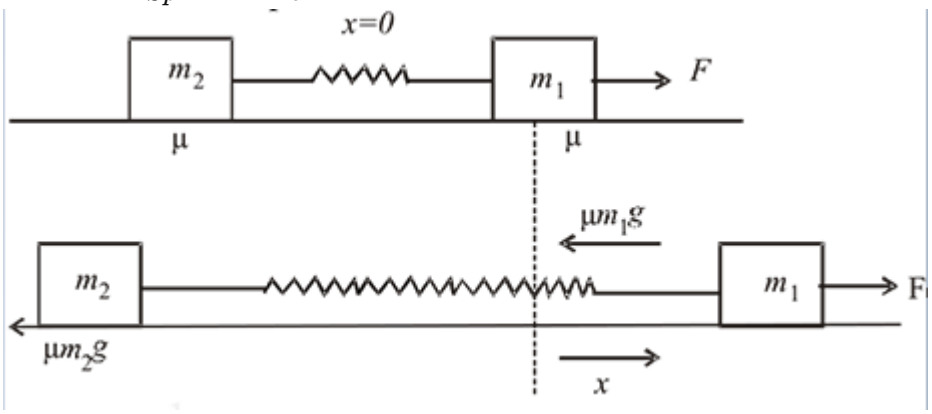
$0 = \left(\frac{\mathrm{v}}{3}\right)^2 - 2\mu\mathrm{gd} \Rightarrow \mathrm{v} = 3\sqrt{2\mu\mathrm{gd}}$

Q.13 Answer:

$\mu m_1 g + \frac{\mu m_2 g}{2}$

Solution:

$W_F + W_{Sp} + W_{\text{fric}} = \Delta k$



$\Rightarrow Fx - \frac{1}{2}kx^2 - \mu m_1 g x = 0$ **and** $kx = \mu m_2 g$

$\Rightarrow F - \frac{1}{2}\mu m_2 g - \mu m_1 g = 0 \Rightarrow F = \mu m_1 g + \frac{\mu m_2 g}{2}$

Q.14 Answer:

$3.33\ \mathrm{m/s}$

Solution:

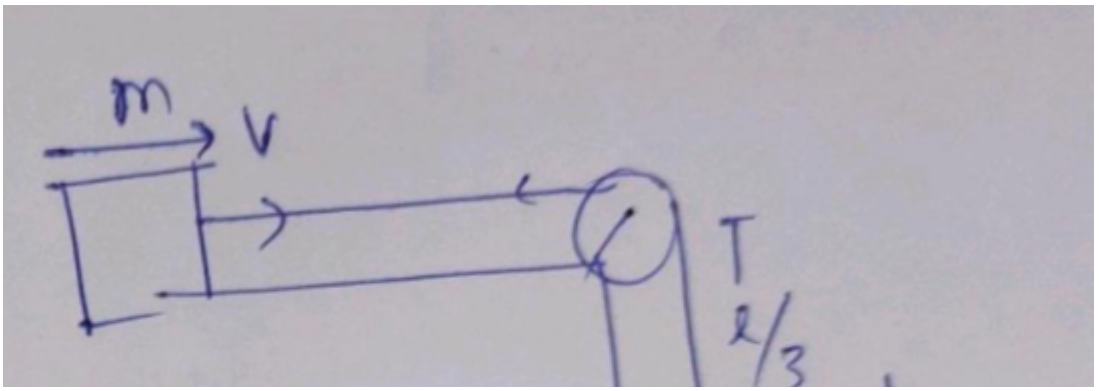
When 'm' descends by $(l/3)$

$\Delta PE = mg\left(\frac{l}{3}\right)$

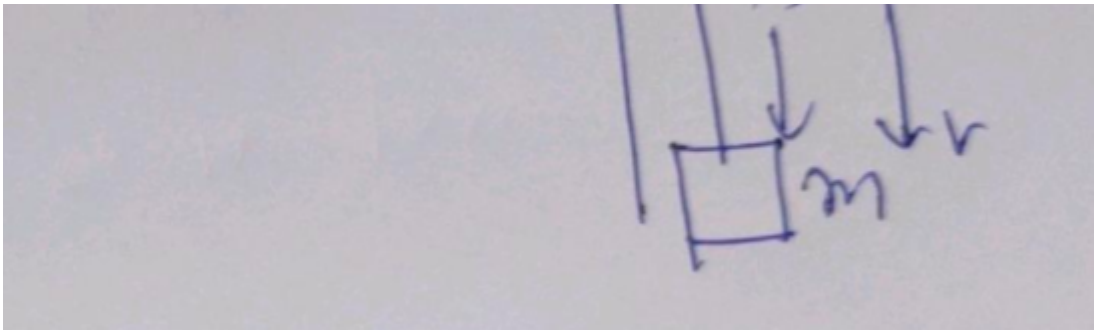
Now

using COE

$\Delta PE = \Delta KE$



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$\frac{\not{m}gl}{3} = \frac{\not{m}v^2}{2} + \frac{\not{m}v^2}{2}$
$\frac{gl}{3} = v^2 \Rightarrow v = \sqrt{\frac{gl}{3}}$
$v = \sqrt{\frac{10\times10}{3\times3}}\ \mathrm{m/s}$
$= \frac{10}{3} m/S$
$= 3.33\ \mathrm{m/sec}$

Q.15 Answer:

$\cos^{-1}\left\{1 - \mathrm{e}^2\left(1 - \cos\theta_0\right)\right\}$

Solution:

Speed just before collision
$= \sqrt{2g\ell\left(1 - \cos\theta_0\right)}$
Speed just after collision $= e\sqrt{2g\ell\left(1 - \cos\theta_0\right)}$
From conservation of energy, $\frac{1}{2}me^2\left[2g\ell\left(1 - \cos\theta_0\right)\right] = mg\ell(1 - \cos\theta)$
$\therefore \theta = \cos^{-1}\left\{1 - e^2\left(1 - \cos\theta_0\right)\right\}$

Q.16 Answer:

$0.1\ \mathrm{m/s^2}$

Solution:

Given: Mass of particle, $\mathrm{M} = 10\ \mathrm{g} = \frac{10}{1000}\ \mathrm{kg}$
radius of circle $\mathrm{R} = 6.4\ \mathrm{cm}$ **Kinetic energy** E **of particle** $= 8 \times 10^{-4}\ \mathrm{J}$ **acceleration** $\mathrm{a_t} = ?$ $\frac{1}{2}\mathrm{mv^2} = \mathrm{E} \Rightarrow \frac{1}{2}\left(\frac{10}{1000}\right)v^2 = 8 \times 10^{-4}$
$\Rightarrow \mathrm{v^2} = 16 \times 10^{-2} \Rightarrow \mathrm{v} = 4 \times 10^{-1} = 0.4\ \mathrm{m/s}$
Now, using $v^2 = u^2 + 2a_t s \quad (s = 4\pi R)$
$(0.4)^2 = 0^2 + 2\mathrm{a_t}\left(4 \times \frac{22}{7} \times \frac{6.4}{100}\right)$
$\Rightarrow \quad a_t = (0.4)^2 \times \frac{7\times100}{8\times22\times6.4} = 0.1\ \mathrm{m/s^2}$

Q.17 Answer:

$\frac{\mathrm{u}}{12}(-\hat{\mathrm{i}} - \sqrt{3}\hat{\mathrm{j}})$

Solution:

From the law of conservation of momentum we know that, $\mathrm{m_1u_1 + m_2u_2 + \ldots = m_1v_1 + m_2v_2 +}$

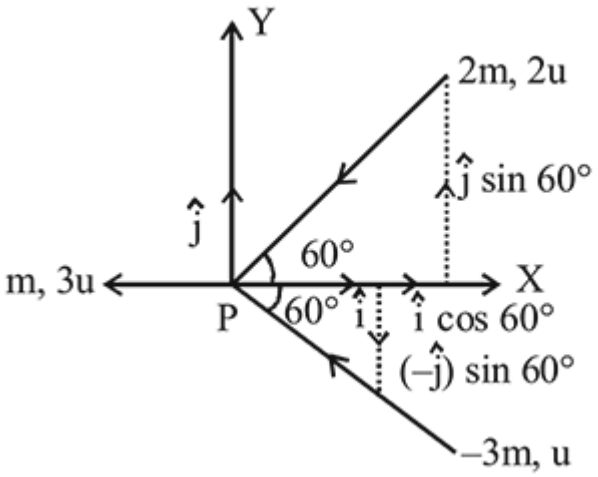
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. . .

Given $m_1 = m, m_2 = 2\ m$ **and** $m_3 = 3\ m$

and $u_1 = 3u, u_2 = 2u$ **and** $u_3 = u$

Let the velocity when they stick $= \vec{v}$ **Then, according to question,**



$$m \times 3u(\hat{i}) + 2\ m \times 2u\left(-\hat{i}\cos 60° - \hat{j}\sin 60°\right)$$

$$3\ m \times u\left(-\hat{i}\cos 60° + \hat{j}\sin 60°\right) = (m + 2\ m + 3\ m)\vec{v}$$

$$3mu\hat{i} - 4mu\frac{\hat{i}}{2} - 4mu\left(\frac{\sqrt{3}}{2}\hat{j}\right) - 3mu\frac{\hat{i}}{2}$$

$$\Rightarrow \quad +3mu\left(\frac{\sqrt{3}}{2}\hat{j}\right) = 6m\vec{v}$$

$$\Rightarrow \quad mu\hat{i} - \tfrac{3}{2}m\hat{i} - \tfrac{\sqrt{3}}{2}mu\hat{j} = 6m\vec{v}$$

$$\Rightarrow \quad -\tfrac{1}{2}mu\hat{i} - \tfrac{\sqrt{3}}{2}mu\hat{j} = 6m\vec{v}$$

$$\Rightarrow \vec{v} = \tfrac{u}{12}(-\hat{i} - \sqrt{3}\hat{j})$$

Q.18 Answer:

$\frac{4\pi r}{u}$

Solution:

If just after collision, relative velocity $= v$ **then** $\frac{v}{u} = \frac{1}{2} \quad \therefore \omega_{rel} = \frac{v}{r} = \frac{u}{2r}$

$\therefore$ **time between** 1st **and** 2nd **collision,** $t = \frac{2\pi}{\omega_{rel}} = \frac{4\pi r}{u}$

Q.19 Answer:

10 cm

Solution:

Let x_A **and** x_B **be the position of ends** A **and** B **at time** t **from the block, then stretched length of the spring will be** $\ell_2 = x_A - x_B$

and so the stretch $\Delta\ell = \ell_2 - \ell_1 = (x_A - x_B) - \ell_1$ (ℓ_1 **natural length of the spring) So,** $U = \frac{1}{2}k\Delta\ell^2 = \frac{1}{2}k\left[(x_A - x_B) - \ell_1\right]^2$

$P = \frac{dU}{dt} = \frac{1}{2}k \cdot 2\left(x_A - x_B - \ell_1\right)\left(\frac{dx_A}{dt} - \frac{dx_B}{dt}\right)$

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$P = F\left(v_A - v_B\right) \quad F = \frac{P}{v_A - v_B}$
$\Delta\ell = \frac{F}{k} = \frac{P}{(v_A - v_B)k} = \frac{20}{(4-2)\times 100}$
$\Delta\ell = 0.1\ \mathrm{m} = 10\ \mathrm{cm}$

Q.20 Answer:

$\mathrm{V}^1 \sin\theta' = \mathrm{V}\sin\theta$

Solution:

As the floor exerts a force on the ball along the normal, & no force parallel to the surface, therefore the velocity component along the parallel to the floor remains constant. Hence $V\sin\theta = V^1\sin\theta^1$

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